COM5231 · Introduction to Wireless and Ad Hoc Networking

Part 4 — Exam Pattern Analysis

All five past papers analysed

SUPPLEMENTARY content not found in the uploaded lecture notes

NOTES ERROR a mistake in the lecture notes, with the verified correction

EXAM how the topic is or could be examined

★ high-priority, frequently examined content

Exam preparation pack, built from the uploaded lecture materials and past papers

Part 4 — Examination Pattern Analysis

This part analyses **all five past assessments** (450 marks in total). Every statement in sections 4.1–4.6 is **counted from the papers you uploaded**. Section 4.7 is clearly labelled as **prediction**. Nobody can know what will appear in your exam, but the patterns below are strong.

Topic numbers: T1 Transmission fundamentals · T2 Wireless channel · T3 Channel correction/capacity · T4 Signal encoding · T5 Multiple access · T6 Spread spectrum · T7 OFDM/OFDMA · T8 Error detection/correction · T9 WLAN 802.11 · T10 Bluetooth/802.15 · T11 Routing/MANET.

4.1 Question-by-question extraction

Exam paper (green; title/year not visible; pages 2–4; 100 marks)

Q	Question (abridged)	Marks	Topic	Type
1(a)	Differences between path loss, multipath propagation, interference	6	T2	Comparison
1(b)	Spread spectrum principles of (i) FHSS (ii) DSSS	4	T6	Explanation
1(c)	Benefits of OFDMA approaches: adjacent, regularly spaced, randomly spaced subcarriers	6	T7	Explanation
1(d)	Describe CDMA; why difficult to jam	4	T5/T6	Explanation
2(a)	Hamming code for 10101111: check bits + codeword	4	T8	**Calculation**
2(b)	CRC: 101110, generator 1001	4	T8	**Calculation**
2(c)	Internet checksum of 1001 1100 1010 0011 (4-bit)	2	T8	**Calculation**
2(d)	Why is ad hoc routing complicated? Three MANET routing categories + examples	10	T11	Essay
3(a)	Compare and contrast link-state and distance-vector	4	T11	Comparison
3(b)	Essential difference: Dijkstra vs Bellman-Ford	4	T11	Comparison
3(c)(i)	Dijkstra on Figure 1 from **t**	4	T11	**Algorithm**
3(c)(ii)	Dijkstra on Figure 1 from **w**	4	T11	**Algorithm**
3(d)	Count-to-infinity; does it occur on link-cost decrease? On connecting two unlinked nodes?	4	T11	Explanation
4(a)	Bluetooth master vs 802.11 base station	4	T10	Comparison
4(b)	Does the hidden terminal problem occur in Bluetooth?	4	T10	Application
4(c)	How can profiles be an advantage for applications?	4	T10	Explanation
4(d)	Can one device be controller of two piconets?	4	T10	Application
4(e)	Scatternet of 2 piconets: collision probability (80 freq.)	4	T10	**Calculation**
5(a)	Mobile node moving away: two BS actions	2	T3	Application
5(b)	Figure 2: hidden node; A and C send simultaneously	4	T9	Scenario
5(c)	Figure 2: CSMA/CA when A sends	4	T9	Scenario
5(d)	Why ACKs in 802.11 but not wired Ethernet?	2	T9	Explanation

Q	Question (abridged)	Marks	Topic	Type
5(e)(i)	Two ISPs' APs both on channel 11	4	T9	Scenario
5(e)(ii)	Channels 1 and 11	4	T9	Scenario

Test 1, version A (T1A; 100 marks)

Q	Question	Marks	Topic	Type
1	Three basic propagation mechanisms in a mobile system	12	T2	Explanation
2	Main problems of signal propagation; why not straight lines; reflection useful and harmful	12	T2	Essay
3	Path loss vs multipath vs interference	9	T2	Comparison
4	FHSS and DSSS principles	10	T6	Explanation
5	Benefits of the three OFDMA approaches	9	T7	Explanation
6	CDMA; why difficult to jam	4	T5/T6	Explanation
7	Mobile node moving away: two BS actions	4	T3	Application
8	Figure 2: hidden node; A and C simultaneous	4	T9	Scenario
9	Figure 2: CSMA/CA when A sends	4	T9	Scenario
10	Why ACKs in 802.11 but not Ethernet	4	T9	Explanation
11(i)	Two ISPs on channel 11	4	T9	Scenario
11(ii)	Channels 1 and 11	4	T9	Scenario
12	Receiving site: occasional loss of signal due to interference; steps to correct	10	T2/T9	Scenario/essay
13	Why many WiFi networks have high error rate from co-channel interference	4	T9	Explanation
14	850 MHz, 125 mW, received 1 µW: distance?	6	T2	**Calculation**

COM5231 Test 1, version B (T1B; 100 marks)

Q	Question	Marks	Topic	Type
1	Single-bit vs burst errors; extent parity is appropriate	10	T8	Explanation
2(a)	CRC: 10011010, $x^3 + x^2 + 1$ (polynomial long division)	10	T8	**Calculation**
2(b)	Leftmost bit inverted: receiver's CRC result; how it knows	6	T8	**Calculation**
3	Three basic propagation mechanisms	12	T2	Explanation
4	Path loss vs multipath vs interference	12	T2	Comparison
5	Three methods to increase capacity of a wireless system	12	T3	Essay
6	FHSS and DSSS principles	10	T6	Explanation
7	CDMA; why difficult to jam	10	T5/T6	Explanation
8	Difference between OFDM and OFDMA	10	T7	Comparison
9	SU-MIMO vs MU-MIMO	8	T3	Comparison

Test 2 2025 (T2-2025; 100 marks)

Q	Question	Marks	Topic	Type
1	Three categories of MANET routing protocols	12	T11	Explanation
2	Dijkstra on Figure 1 from **x**	8	T11	**Algorithm**
3	Count-to-infinity problem	5	T11	Explanation
4	Hidden terminal problem; does it occur in Bluetooth?	10	T9/T10	Explanation
5	Compare wired and wireless LANs; unique WLAN design concerns	10	T9	Comparison/essay
6	Receiving site with interference: steps to correct	10	T2/T9	Scenario
7	Bluetooth profiles: benefit to applications	10	T10	Explanation
8	Scatternet collision probability	10	T10	**Calculation**
9	One device as controller of two piconets?	5	T10	Application
10(i)	Two ISPs on channel 11	10	T9	Scenario
10(ii)	Channels 1 and 11	10	T9	Scenario

COM5231 Test 2 2026 (T2-2026; 50 marks)

Q	Question	Marks	Topic	Type
1	Routing vs packet forwarding vs switching: purpose and decision information	6	T11	Comparison
2(a)–(c)	Hamming for 11001010 (inequality, placement, codeword)	8	T8	**Calculation**
3(a)–(c)	CRC: 110101, G = 1011	6	T8	**Calculation**
4(a)–(b)	Checksum 1010 1100 0110 1001 + receiver result	4	T8	**Calculation**
5	Define single-bit and burst error; burst length	3	T8	Definition
6(a)–(c)	Even parity: single-bit? burst? can it correct?	5	T8	Application
7	Two ways profiles benefit developers/users	4	T10	Explanation
8	Bluetooth central vs 802.11 AP: medium access, connecting to other networks	4	T10	Comparison
9(a)–(b)	Dijkstra from A (new figure); final costs and routes	10	T11	**Algorithm**

4.2 Marks by topic (counted)

Topic	Exam	T1A	T1B	T2-2025	T2-2026	**Total**	**% of 450**
T2 Wireless channel (incl. interference troubleshooting, path-loss calc)	6	49	24	10	—	**89**	**19.8%**
T9 WLAN 802.11	18	24	—	35	—	**77**	**17.1%**
T11 Routing/MANET *(no notes)*	30	—	—	25	16	**71**	**15.8%**
T8 Error detection/correction	10	—	26	—	26	**62**	**13.8%**

Topic	Exam	T1A	T1B	T2-2025	T2-2026	Total	% of 450
T10 Bluetooth/802.15	20	—	—	30	8	**58**	**12.9%**
T3 Channel correction/capacity/MIMO	2	4	20	—	—	**26**	**5.8%**
T7 OFDM/OFDMA	6	9	10	—	—	**25**	**5.6%**
T6 Spread spectrum	4	10	10	—	—	**24**	**5.3%**
T5 Multiple access (CDMA)	4	4	10	—	—	**18**	**4.0%**
T1 Transmission fundamentals	—	—	—	—	—	**0**	0%
T4 Signal encoding	—	—	—	—	—	**0**	0%
Total	100	100	100	100	50	**450**	100%

The exam paper itself (the best guide to your final exam): T11 routing **30**, T10 Bluetooth **20**, T9 WLAN **18**, T8 coding **10**, T2 **6**, T7 **6**, T6 **4**, T5 **4**, T3 **2**. The exam **samples every half of the module**: Q1 = Test 1 material (channel, SS, OFDMA, CDMA); Q2–Q3 = coding + routing; Q4 = Bluetooth; Q5 = WLAN.

4.3 Repeated and closely related questions (counted)

#	Question (essentially identical wording)	Papers	Count
R1	**Bluetooth profiles add complexity; how are they an advantage to applications?**	Exam 4(c), T2-2025 Q7, T2-2026 Q7, BT exercise 19	**4**
R2	**Path loss vs multipath vs interference**	Exam 1(a), T1A Q3, T1B Q4	**3**
R3	**Spread spectrum principles of FHSS and DSSS**	Exam 1(b), T1A Q4, T1B Q6	**3**
R4	**Describe CDMA; why difficult to jam**	Exam 1(d), T1A Q6, T1B Q7	**3**
R5	**Two ISPs' APs on channel 11; then channels 1 and 11**	Exam 5(e), T1A Q11, T2-2025 Q10	**3**
R6	**CRC calculation** (different numbers)	Exam 2(b), T1B Q2, T2-2026 Q3 (+ worksheet)	**3**
R7	**One device as controller/master of two piconets**	Exam 4(d), T2-2025 Q9, BT exercise 9	**3**
R8	**Scatternet collision probability, 80 frequencies**	Exam 4(e), T2-2025 Q8, BT exercise 20	**3**
R9	**Bluetooth master vs 802.11 base station / AP**	Exam 4(a), T2-2026 Q8, BT exercise 8	**3**
R10	**Three basic propagation mechanisms**	T1A Q1, T1B Q3 (+ lecturer review Q)	**2**
R11	**Benefits of OFDMA approaches (adjacent / regular / random)**	Exam 1(c), T1A Q5 (+ OFDM review 8.10)	**2**
R12	**Hidden node with Figure 2; A and C transmit simultaneously**	Exam 5(b), T1A Q8	**2**
R13	**CSMA/CA with Figure 2, A sending**	Exam 5(c), T1A Q9	**2**
R14	**Why ACKs in 802.11 but not in Ethernet**	Exam 5(d), T1A Q10	**2**
R15	**Mobile node moving away: two actions**	Exam 5(a), T1A Q7	**2**

#	Question (essentially identical wording)	Papers	Count
R16	**Interference at a receiving site: steps to correct**	T1A Q12, T2-2025 Q6	**2**
R17	**Hamming code** (different data)	Exam 2(a), T2-2026 Q2 (+ worksheet)	**2**
R18	**Internet checksum, 4-bit words**	Exam 2(c), T2-2026 Q4 (+ worksheet)	**2**
R19	**Single-bit vs burst errors; parity**	T1B Q1, T2-2026 Q5–6 (+ worksheet)	**2**
R20	**Three categories of MANET routing**	Exam 2(d), T2-2025 Q1	**2**
R21	**Dijkstra on the same Figure 1** (from x, t, w)	Exam 3(c), T2-2025 Q2	**2** (+ T2-2026 Q9 new graph)
R22	**Count-to-infinity**	Exam 3(d), T2-2025 Q3	**2**
R23	**Hidden terminal problem in Bluetooth**	Exam 4(b), T2-2025 Q4	**2**

Key finding: these **23 question types account for the large majority of the 450 marks.** Of the 24 sub-questions in the exam paper, **22** also appear (same or nearly the same wording) in a test, the worksheet or the Bluetooth exercise sheet. The only exam-only items are Q3(a) link-state vs distance-vector and Q3(b) Dijkstra vs Bellman-Ford. The lecturer **reuses questions heavily**, often word for word, and changes only the numbers in calculations.

Questions copied word for word from sources:

- **Test 1A Q14** = *Wireless Communication Problems*, **Problem 2.7** (same numbers).
- **Exam 2(a)** (Hamming 10101111) and **Exam 2(c)** (checksum 1001 1100 1010 0011) = the **error-detection worksheet** exercises.
- **Exam Q4(a), (c), (d), (e)** = **Bluetooth exercises 8, 19, 9, 20**.
- **T1B Q1** = the last worksheet question; **T1B Q2** = the worksheet's CRC example with a different message.
- **Exam Q3 / T2-2025 Q2** Figure 1 = Kurose & Ross Ch. 5 network; **T2-2026 Q9** = Tanenbaum's Dijkstra figure.
- **T1B Q9, T1A Q1** = the lecturer's own review list on *The Wireless Channel* slide 72.

4.4 Common question formats

Format	Typical wording	Typical marks	Examples
Compare / differences	"What are the differences...", "Compare and contrast...", "Distinguish..."	4–12	R2, R9, OFDM vs OFDMA, SU vs MU-MIMO, LS vs DV, wired vs wireless LAN
Explain principle	"Explain the principles of...", "Describe..."	4–12	FHSS/DSSS, CDMA, propagation mechanisms
Scenario / figure	"Use Figure 2 to...", "Suppose two ISPs..."	2–10	Hidden node, CSMA/CA, ISP café, interference site
Yes/No + justify	"Does ... occur? Explain", "Is there any reason why..."	4–10	Hidden terminal in BT; two piconets; count-to-infinity

Format	Typical wording	Typical marks	Examples
Calculation	"Show...", "What is the value..."	2–16	Hamming, CRC, checksum, path loss/distance, collision probability
Algorithm trace	"Apply Dijkstra... step-by-step"	4–10	Figure 1, T2-2026 figure
Essay	"Discuss the steps...", "State and explain the three..."	10–12	Interference troubleshooting, capacity methods, MANET categories

4.5 Calculation and algorithm questions (counted)

Calculation	Appears	Marks available
CRC	Exam, T1B, T2-2026	4 + 16 + 6 = **26**
Dijkstra	Exam (×2), T2-2025, T2-2026	8 + 8 + 10 = **26**
Hamming	Exam, T2-2026	4 + 8 = **12**
Scatternet probability	Exam, T2-2025	4 + 10 = **14**
Internet checksum	Exam, T2-2026	2 + 4 = **6**
Free-space distance/power	T1A	**6**

≈ 90 of 450 marks (20%) are pure calculation/algorithm. These are the **most reliable marks**: the method never changes, only the numbers.

4.6 Depth of knowledge required

- Most 4-mark questions want **2–4 precise, explained points** (≈ one point per 1–2 marks).
- 10–12 mark questions want **structure**: a definition, then 3–5 explained points or a comparison table, then a conclusion.
- Scenario questions reward **applying** the theory (CSMA/CA steps, NAV, CTS) to the named nodes (A, B, C), not reciting it.
- Calculations reward **showing every step** (formula, substitution, each XOR or table row, the final answer with units). Several questions explicitly say *"clearly presenting your step-by-step process"*.
- **No questions** in any paper on **Topic 1 (Nyquist/Shannon)** or **Topic 4 (ASK/FSK/PSK/QAM)**. They are still in the module and the lecturer's exercises, so don't skip them, but rank them lower.

4.7 PREDICTION (not fact): priorities for the final exam

These are **predictions** based on the patterns above. They are not knowledge of the exam.

Very high probability (appeared in the exam AND in tests; prepare word-perfect answers):

1. Bluetooth profiles advantage (R1); master vs AP (R9); two piconets (R7); scatternet probability (R8); hidden terminal in BT (R23).
2. Two-ISP café scenario (R5); hidden node and CSMA/CA with Figure 2 (R12, R13); ACKs (R14).
3. Dijkstra (any graph; Figure 1 from any node); LS vs DV; Dijkstra vs Bellman-Ford; count-to-infinity; MANET categories.
4. Hamming, CRC and checksum with **new numbers**.
5. Path loss vs multipath vs interference; FHSS/DSSS; CDMA jamming; OFDMA approaches.

Medium probability (appeared in tests only): three propagation mechanisms; propagation problems; capacity methods; OFDM vs OFDMA; SU vs MU-MIMO; wired vs wireless LAN; interference troubleshooting; co-channel interference; routing/forwarding/switching; parity and burst errors; path-loss distance calculation.

Lower probability but examinable (in notes/review questions, never yet examined): Nyquist/Shannon calculations; modulation (ASK/FSK/PSK/QAM); FDMA/TDMA details and formulas; cellular generations; SC-FDMA; 802.15.3/.4 and ZigBee; Bluetooth packet format and links; PCF/IFS details; channel correction mechanisms (FEC, equalization, AMC).

Recommended study time split (prediction-based): Routing 20% · WLAN 20% · Bluetooth 20% · Error coding 15% · Wireless channel/SS/CDMA/OFDMA 20% · Everything else 5%.